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Adversarial Robustness and Gradient Masking Defenses in Deep Neural Network Classifiers

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ABSTRACT

Deep learning models deployed in threat detection remain vulnerable to adversarial perturbations that evade detection with imperceptible noise. We formulate an adversarial defense mechanism combining stochastic activation pruning with certified robust optimization bounds. Our approach eliminates obfuscated gradient masking by certifying lower bounds on the minimum perturbation radius required to alter classification logits. On adversarial robustness benchmarks (AutoAttack and PGD-100), our defended model maintains 74.2% robust accuracy against adaptive white-box attacks. However, certified defense training incurs a 10x computational slowdown during backward passes and introduces a 3.1% degradation in clean classification accuracy.

Keywords— Cybersecurity, Adversarial Robustness, Gradient Masking, Neural Networks

INTRODUCTION

Malware classifiers and intrusion detection systems powered by deep learning are increasingly targeted by adversarial evasion attacks. Many proposed defenses merely mask gradients, giving an illusion of security that collapses under adaptive optimization attacks.

METHODOLOGY

We introduce randomized smoothing with certified l2-norm guarantees directly into the loss objective. During backpropagation, Lipschitz regularization bounds the gradient norm, preventing adversarial gradient-free exploitation.

RESULTS

Against white-box PGD attacks with epsilon=8/255, our model maintained 74.2% adversarial accuracy, whereas undefended models dropped to 0% and gradient-masking defenses fell to 12.4%.

LIMITATIONS

Robust training requires multiple stochastic forward-backward passes per iteration, increasing training time by an order of magnitude and slightly penalizing classification accuracy on non-adversarial samples.

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